UNIVERSITY OF TEXAS ARLINGTON
COLLEGE OF SCIENCE

Advanced Classical Mechanics

Lecture 4: Newtonian Statics

Readings: Morin - Chapter 2

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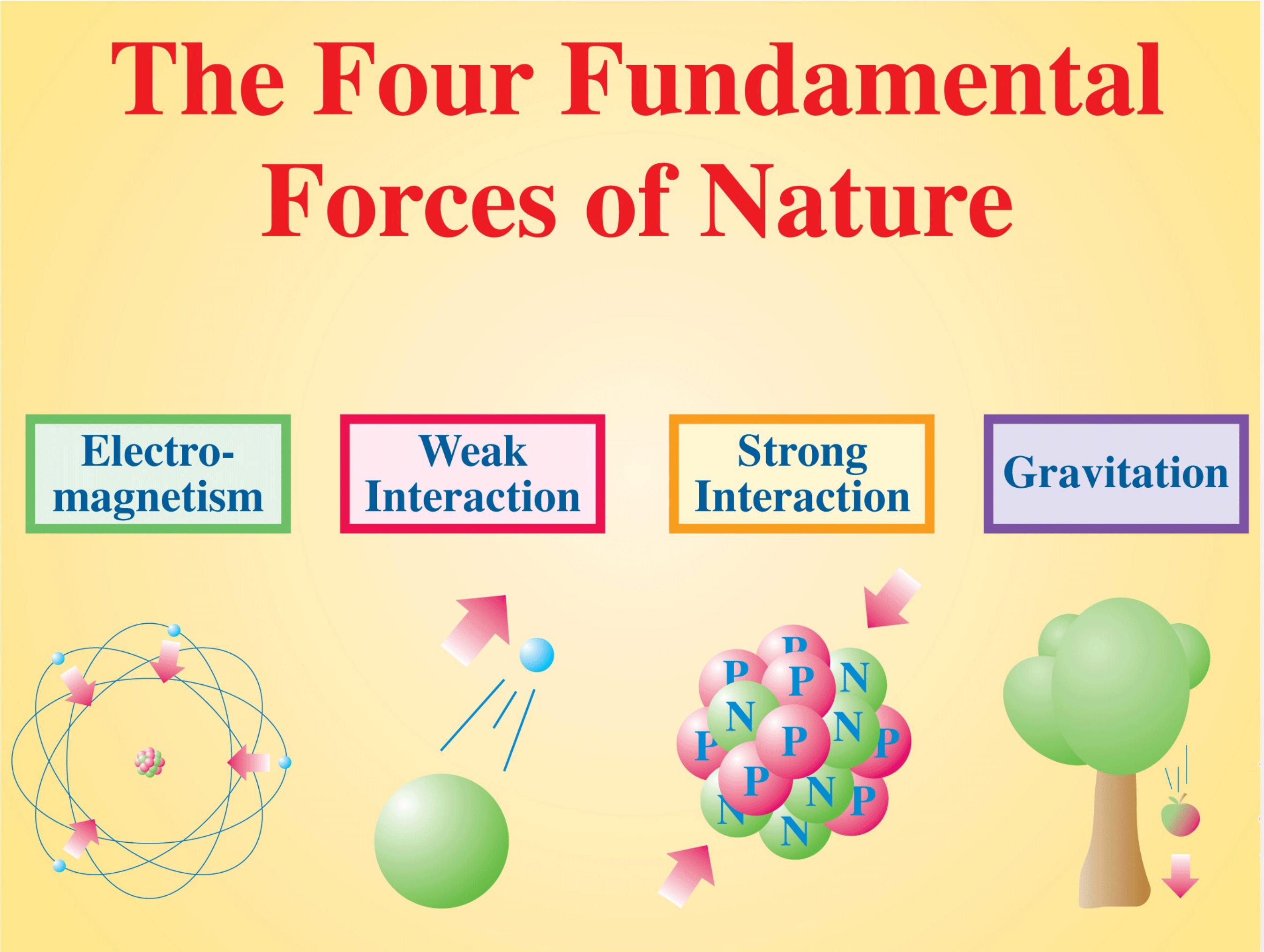
Newtonian Mechanics (by Isaac Newton at age 44 in 1687)



Newtonian mechanics primarily deals with **Gravitational force** and **electromagnetic reactions** to the gravitational force.

Newtonian Mechanics (by Isaac Newton at age 44 in 1687)

Newton's axioms:

1. **Principle of Inertia.** A body remains at rest or in motion at a constant speed in a straight line unless a force acts upon it.

    - It is impossible to determine which inertial frames are stationary or moving at constant speed.

    - Einstein's theory of relativity extends this notion of relativity in motion to relativity in space & time.

2. $\boldsymbol{F} = m\boldsymbol{a}$. At any instant of time, the net force on a body is equal to the body's acceleration multiplied by its mass ($\boldsymbol{F} = m\boldsymbol{a}$). Newton originally meant $\boldsymbol{F} = \mathrm{d}\boldsymbol{P}/\mathrm{d}t$ where $\boldsymbol{P}$ is the object's momentum.

3. **Action equals reaction.** The forces exerted by two bodies on each other are equal in magnitude but in opposite directions.

    - Newton's third law is a special case of the more general **law of momentum conservation**.

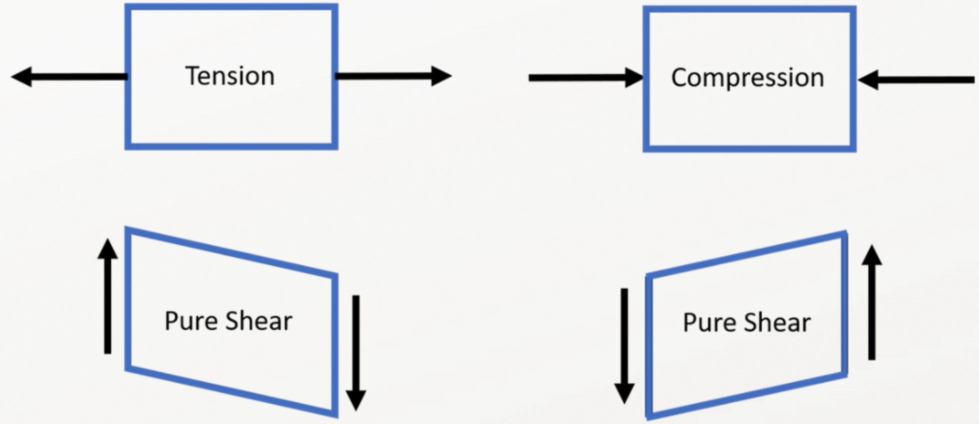
$$\boldsymbol{P} = \boldsymbol{P}_1 + \boldsymbol{P}_2$$

Static system.

- A system that is stationary (no motions) with respect to the reference frame.
- The net force acting on a static system is zero (Recall Newton's second law): $\sum \boldsymbol{F}_i = m\boldsymbol{a} = 0$
- The net torque acting on a static system is zero: $\sum \boldsymbol{\tau}_i = 0$
- **Question.** Does a net zero force on a system imply that the system is in a static setup?

The many faces of the electromagnetic force in classical mechanics

- **Tension force**



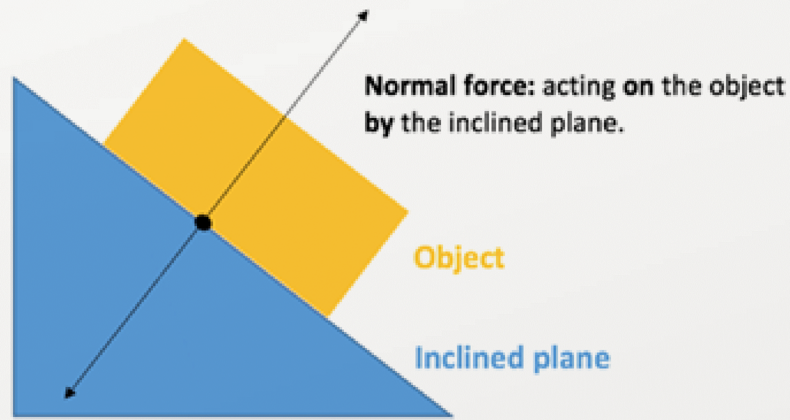
Tension is the stretching force that is transmitted axially along (usually an elongated) object (e.g., rope).

  - For a stationary object, tension is felt equally and in opposite directions at each point within the object (Why?).

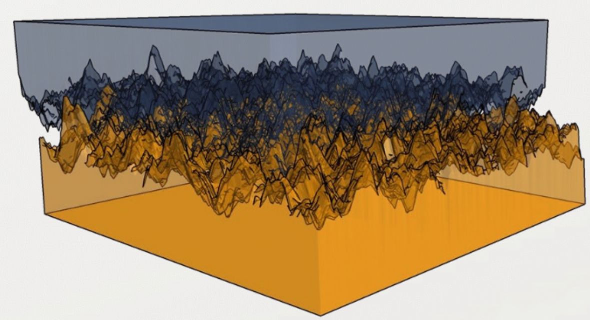
- **Contact forces**

Forces that originate from the contact of two objects with each other.

1. **Normal force.** The force an object exerts perpendicular to its contact surface on another object. Normal force typically arises in response to a **compression force**, also known as "**compressive tension**" or "**negative tension**" (e.g., gravity).

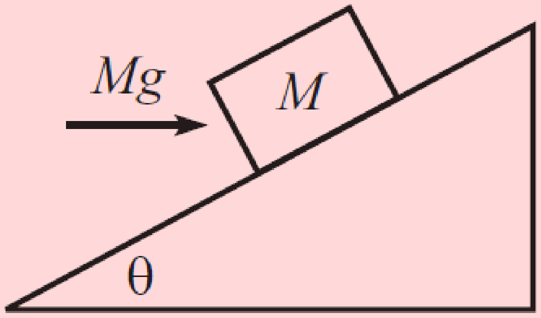


2. **(Dry) Friction.** The force an object exerts parallel to its contact surface on another object. The Coulomb friction frequently quantifies the dry friction equation, as $F_f = \mu F_N$ where μ is the **coefficient of friction** and F_N is a normal force.



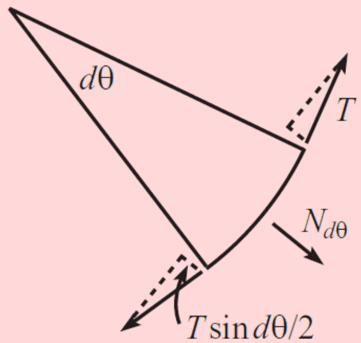
Newtonian Static: Balancing forces and torques

Problem 1. A block of mass M rests on a fixed plane inclined at an angle θ. You apply a horizontal force of Mg on the block, as shown in the figure. Assume that the friction force between the block and the plane is large enough to keep the block at rest.



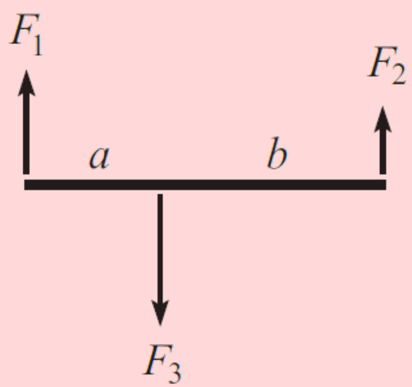
a) What are the normal and friction forces (call them N and F_f) that the plane exerts on the block?
b) If the coefficient of static friction is μ, for what range of angles θ will the block in fact remain at rest?

Problem 2. A rope wraps an angle θ around a pole. You grab one end and pull with a tension T_0. The other end is attached to a large object, say, a boat.



If the coefficient of static friction between the rope and the pole is μ, what is the largest force the rope can exert on the boat if the rope is not to slip around the pole?

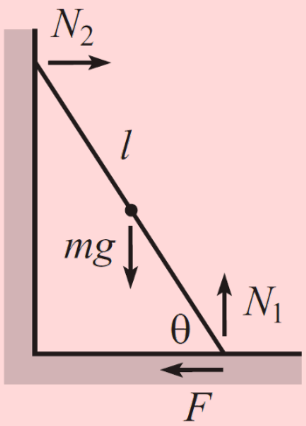
Problem 3. Balancing torques. Consider the following static setup where three forces are applied perpendicular to a stick. $\boldsymbol{F_1}$ and $\boldsymbol{F_2}$ are the forces at the ends, and $\boldsymbol{F_3}$ is the force in the interior. For the system to be static $\boldsymbol{F_3} = \boldsymbol{F_1} + \boldsymbol{F_2}$, because the stick is at rest. Show that for this system to be static, the following relation must also hold:



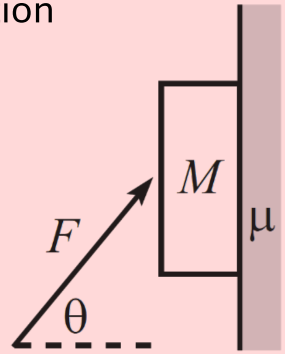
$$\boldsymbol{F_3} a = \boldsymbol{F_2}(a + b)$$

Newtonian Static: Balancing forces and torques

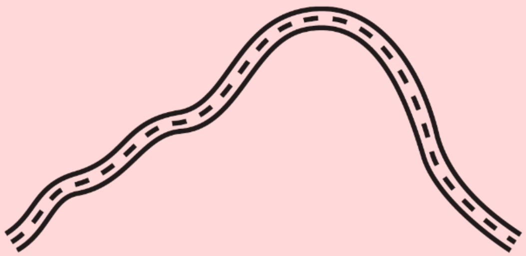
Problem 4. A ladder leans against a frictionless wall. If the coefficient of friction with the ground is μ, what is the smallest angle the ladder can make with the ground and not slip?



Problem 5. A book of mass M is positioned against a vertical wall. The coefficient of friction between the book and the wall is μ. You wish to keep the book from falling by pushing on it with force $\boldsymbol{F}$ applied at an angle θ with respect to the horizontal ($-\pi/2 < \theta < \pi/2$).
a) For a given θ, what is the minimum F required?
b) For what θ is this minimum F the smallest? What is the corresponding minimum F?
c) What is the limiting value of θ below which there does not exist an F that keeps the book up?

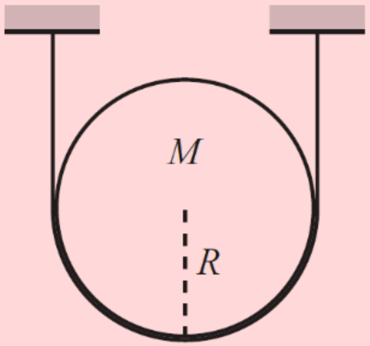


Problem 6. Motionless chain. A frictionless tube lies in the vertical plane and is in the shape of a function that has its endpoints at the same height but is otherwise arbitrary. As shown in the figure, a chain with uniform mass per unit length lies in the tube from end to end Show, by considering the net force of gravity along the curve, that the chain doesn't move.



Newtonian Static: Balancing forces and torques

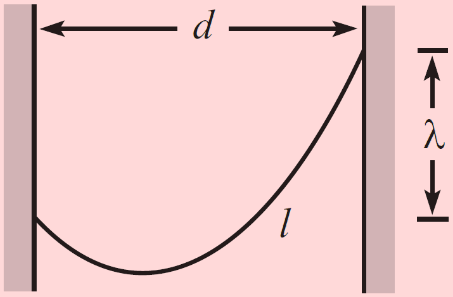
Problem 7.



a) A disk of mass M and radius R is held up by a massless string, as shown in the figure. The surface of the disk is frictionless.
What is the tension in the string?
What is the normal force per unit length that the string applies to the disk?

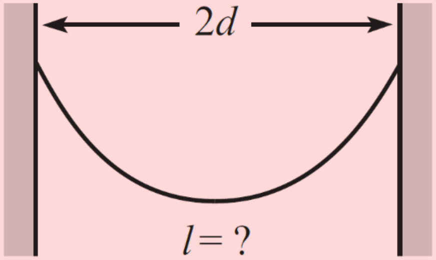
b) Let there now be friction between the disk and the string, with coefficient μ.
What is the smallest possible tension in the string at its lowest point?

Problem 8. Hanging chain.



a) A chain with uniform mass density per unit length hangs between two given points on two walls. Find the general shape of the chain. Aside from an arbitrary additive constant, the function describing the shape should contain one unknown constant.
b) The unknown constant in your answer depends on the horizontal distance d between the walls, the vertical distance λ between the support points, and the length l of the chain, as in the figure. Find an equation involving these given quantities that determines the unknown constant.

Problem 9. Hanging gently.

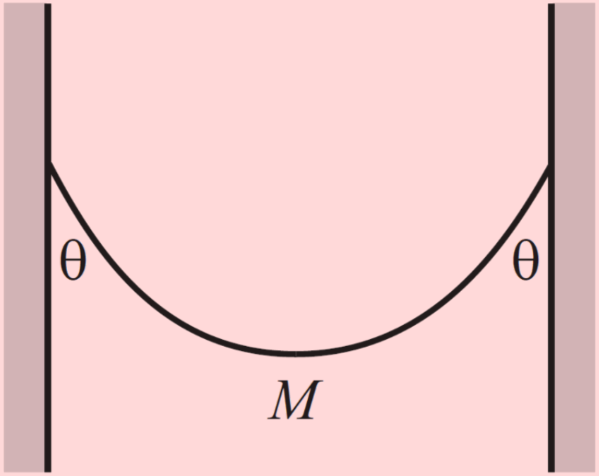


A chain with uniform mass density per unit length hangs between two supports located at the same height, a distance $2d$ apart.
What should the length of the chain be so that the magnitude of the force at the supports is minimized? You may use the fact that a hanging chain takes the form, $y(x) = (1/\alpha)\cosh(\alpha x)$. You will eventually need to solve an equation numerically.

Newtonian Static: Balancing forces and torques

Problem 10. Hanging chain.
A chain with mass M hangs between two walls, with its ends at the same height.
As in the figure, the chain makes an angle θ with each wall.
Find the tension in the chain at the lowest point.
Solve this in two different ways:

a) Consider the forces on half of the chain.
b) Use the fact that the height of a hanging chain is given by $y(x) = (1/\alpha)\cosh(\alpha x)$, and consider the vertical forces on an infinitesimal piece at the bottom. This will give you the tension in terms of α. Then find an expression for α in terms of the given angle θ.



Problem 11. Stacking blocks.
N blocks of length l are stacked on top of each other at the edge of a table, as shown in the figure for $N = 4$. What is the largest horizontal distance the rightmost point on the top block can hang out beyond the table?
How does your answer behave for $N \to \infty$.